

NAME.....SCHOOL.....DATE.....

233/1

CHEMISTRY PAPER 1 TIME: 2 HOURS

1. (a) State one disadvantage of using flower extracts as acid – base indicators. (1mk)

 (b) Name the indicator that can be used in the laboratory to tell the pH of lemon juice.(1mk)

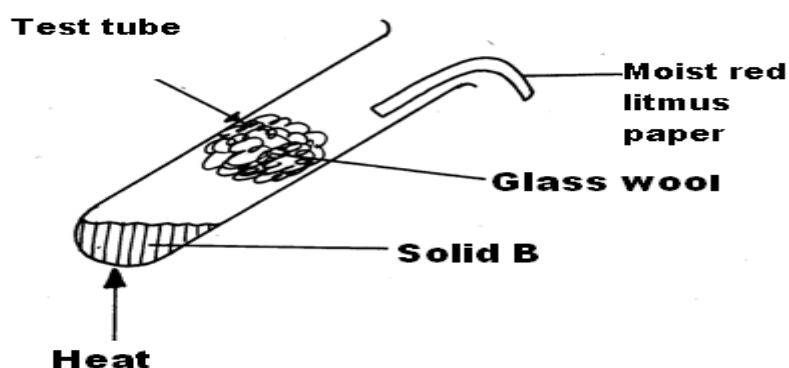
 (c) Differentiate between strong and weak acids. (1mk)

2. (a) What are isotopes? (1mk)

 (b) Determine the number of neutrons in $^{18}_8\text{O}$ (1mk)
3. When magnesium is burnt in air it reacts with both oxygen and nitrogen gas giving a white ash.
 Write two equations for the reactions that take place. (1mk)

4. A solution contains 29.1g per litre of aluminium Sulphate. Calculate the number of Sulphate ions in 350cm³ of the solution.
 (Al = 27, S = 32, O = 16) Avogadro's constant = 6.0×10^{23} . (2mks)

5. When a solid B was heated in a test-tube, it gave off two gases. The two gases were separated by passing them through a plug of glass wool in a test-tube as shown below.



The first gas which evolved turned moist red litmus paper to blue. Later the other gas involved turned the litmus back to red.

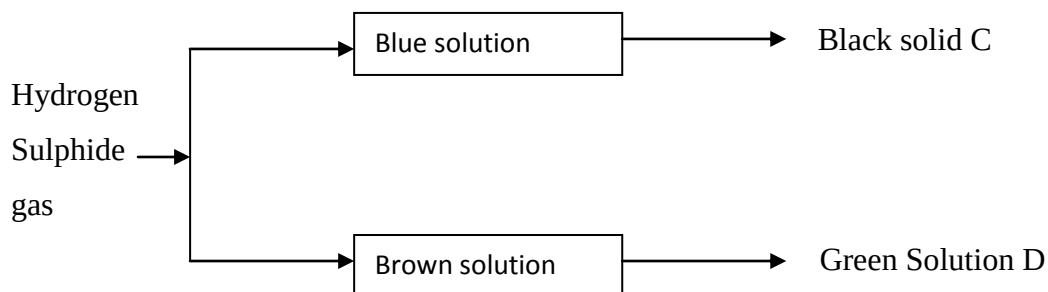
- (a) Identify solid B (1mk)

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- (b) Write the equation for the reaction that take place in the test tube (1mk)

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6. Hydrogen Sulphide gas was bubbled into two solutions of metallic nitrates as shown in the flow diagram below



- (a) Identify the black solid C (1mk)

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- (b) Write an ionic equation for the formation of the green solution (1mk)

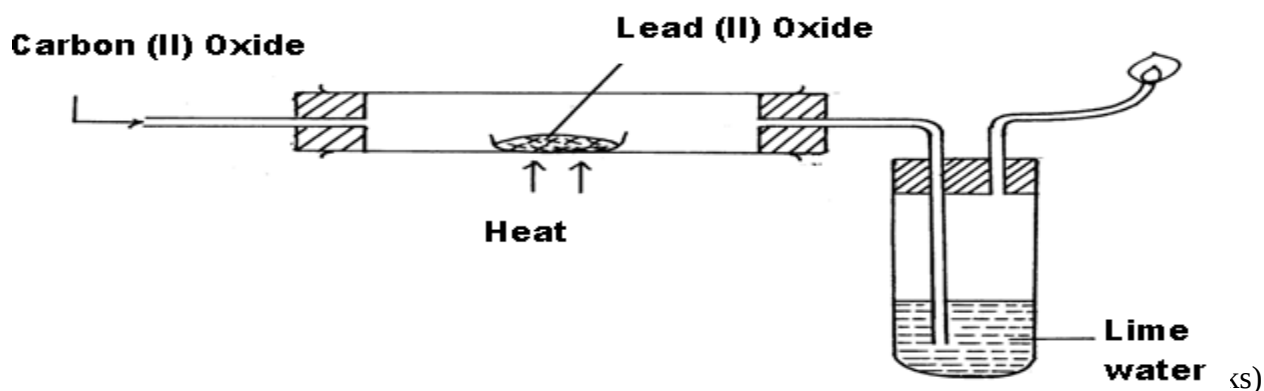
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- (c) State the property of Hydrogen Sulphide shown by the formation of solution D (1mk)

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7. The apparatus shown below was used to investigate the effect of Carbon (II) Oxide on Lead (II) Oxide.



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- (b) Write the equation for the reaction that take place in test-tube E. (1mk)

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8. (a) Lime water is also used to soften hard water. Why is this method not preferred to soften hard water. (1mk)

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- (b) Name a compound that causes temporary hardness of water. (1mk)

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- (c) State one disadvantage of using hard water for domestic purpose. (1mk)

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9. (a) Name the compound below



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- (b) Draw and name other isomers of the compound in (a) above. (2mks)

10. A white solid dissolve in water to form a colourless solution. The colourless solution forms a white precipitate with Ammonia solution but dissolve in excess alkali. The colourless solution forms a white precipitate with Lead (II) Nitrate solution. The white Precipitate dissolve on warming to form a colourless solution.
- (a) Write the chemical formulae for the ion formed when the colourless solution react with excess ammonia solution. (1mk)
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- (b) Write the name of the ion present in the white solid. (1mk)
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- (c) What is an alkali (1mk)
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11. The solubility of Potassium Manganate (VII) at 20°C is 13g per 100g of water and at 90°C is 60g per 100g of water.
- (a) Determine the mass of Pottassium (VII) Manganate present in 80g of saturated solution at 90°C . (1mk)
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- (b) Calculate the mass of Pottassium (VII) Manganate that would crystallize out if the solution in (a) were cooled to 20°C . (2mks)
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12. (a) Explain why concentrated Sulphuric (VI) acid is a poor electrolyte and has no effect on

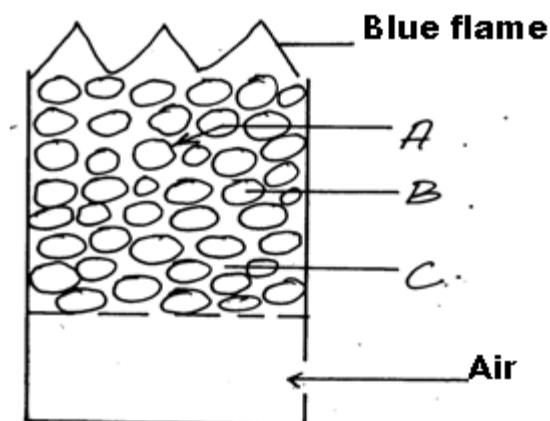
blue litmus paper whereas 2M Sulphuric (VI) acid conducts electricity and changes blue litmus paper red. (1mk)

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(b) What is a binary electrolyte. (1mk)

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13. An element F has a relative atomic mass of 88. When a current of 0.5 amperes was persed through the fused chloride of F for 20 minutes and 20 seconds, 0.278g of F were deposited at the cathode. Determine the charge on ion of F (1Faraday = 96500C). (2mks)

14. Below is a cross- section of a charcoal burner



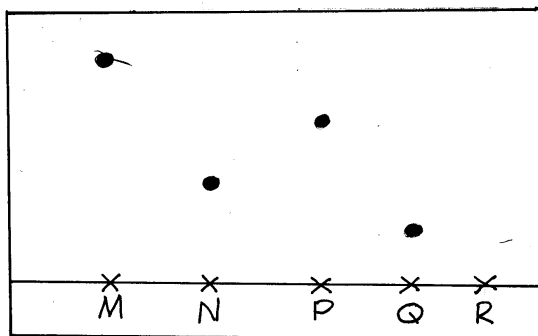
- (a) Charcoal is a form of impure carbon. Name any other two allotropes of carbon. (1mk)

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- (b) Write an equation for the reaction taking place at the part marked B. (1mk)

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15. The diagram below represents paper chromatogram of four types of sugar.



- (a) Identify the most stable sugar. (1mk)

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- (b) On the diagram, show the chromatogram of R. Given that R contains all the other sugars (2mks)

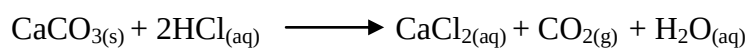
16. The empirical formula of a hydrocarbon is C_2H_3 . The hydrocarbon has a relative molecular Mass of 54. (H = 1, C = 12)

- (a) Determine the molecular formula of the hydrocarbon. (1mk)

- (b) Draw the structural formulae of the hydrocarbon in (a) (1mk)

- (c) To which homologous series does the hydrocarbon in (b) above belong? (1mk)

17. State three factors that increase the rate of reaction for the following reaction. (3mks)



18. In terms of structure and bonding explain the following

(a) Melting point of Magnesium is higher than that of Sodium.

(3mks)

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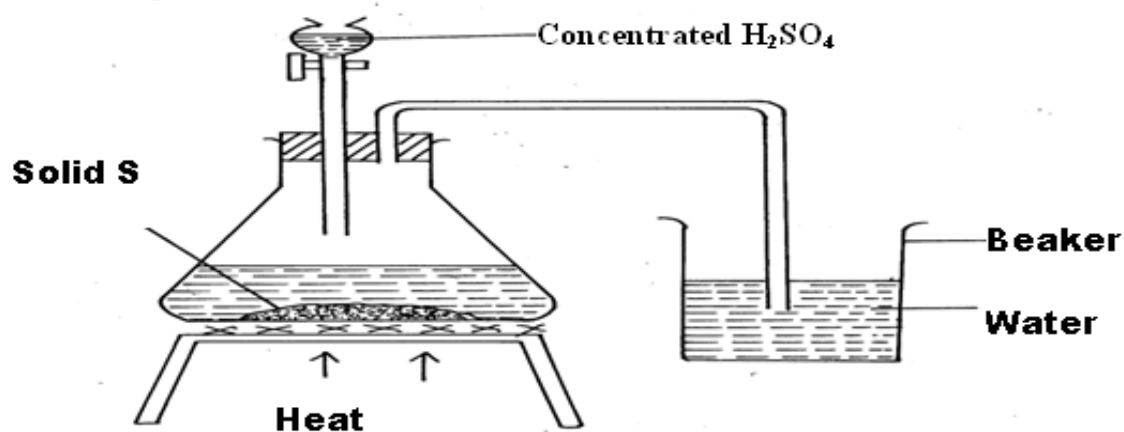
(b) Melting point of Chlorine is lower than that of Iodine.

(1mk)

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19. The set-up below was used to prepare a solution of hydrogen chloride gas



(a) Identify solid S

(1mk)

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(b) Identify one mistake in the set up.

(1mk)

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(c) Write an equation for the reaction taking place in the flask.

(1mk)

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20. (a) State Gay Lussac's law

(1mk)

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- (b) 10cm^3 of a gaseous hydrocarbon, C_xH_y required 30cm^3 of oxygen for complete combustion. If steam and 20cm^3 of Carbon (IV) oxide were produced, what is the value of x in C_xH_y . (2mks)

21. Starting with lead metal, describe how a dry sample of Lead(II) chloride can be prepared in the laboratory. (3mks)

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22. An ion T^{2-} has an electronic arrangement of 2.8

- (a) What is the atomic number of the element. (1mk)

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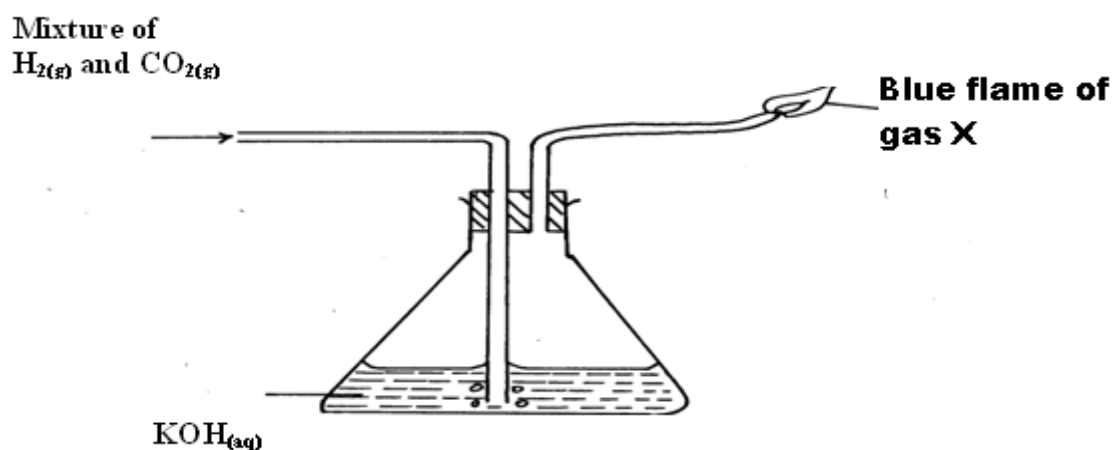
- (b) To which group and period does the element belong to;

Group _____ (1mk)

Period _____ (1mk)

23. Using dot (.) and cross (x) diagram show the type of bond present in hydrogen ion, H_3O^+ (H = 1, O = 8) (2mks)

24. A mixture of Hydrogen gas and Carbon (IV) oxide are passed through Potassium hydroxide solution as shown below.



- (a) State the observation made in the conical flask. (1mk)

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- (b) Write the equation for the reaction that takes place in:-

- (i) the conical flask (1mk)

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- (ii) the burning of gas x (1mk)

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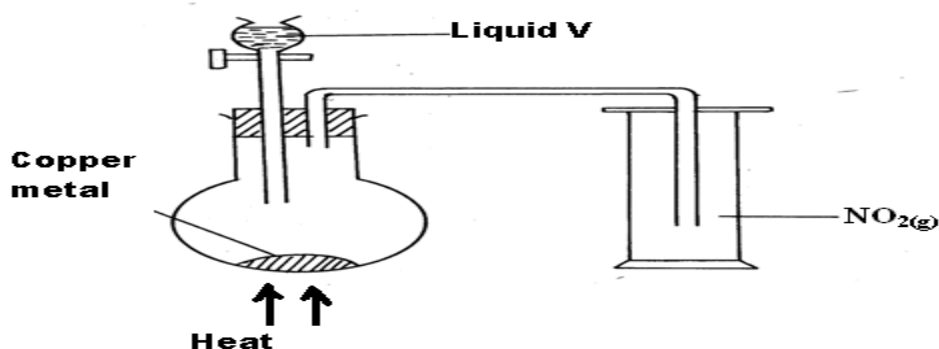
25. 20cm^3 of a solution containing 4g per litre of Sodium hydroxide was neutralized by 8cm^3 of dilute Sulphuric (VI) acid. Calculate the concentration of the acid in moles per litre.

(Na = 23, O = 16, H=1)

(3mks)

26. Given that the hydration energies of $\text{Ca}^{2+}_{(\text{g})}$ and $\text{Cl}^{-}_{(\text{g})}$ are -1562KJ/Mole and -364KJ/Mole respectively. The heat of solution (ΔH_{soln}) for one Mole of CaCl_2 is -82.9KJ/Mole . Determine the lattice energy for CaCl_2 . (2mks)

27. The diagram below is used to prepare nitrogen (IV) oxide gas.



- (a) Identify substance V (1mk)
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- (b) State and explain one precaution taken when carrying out the experiment. (2mks)
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28. Electrode potentials for the half cells are shown below. Use the information to answer the questions that follow.
- $\text{Sn}^{2+}_{(\text{aq})} + 2\text{e}^{-} \longrightarrow \text{Sn}_{(\text{s})}; E^{\theta} = -0.14\text{V}$
- $\text{Cu}^{2+}_{(\text{aq})} + 2\text{e}^{-} \longrightarrow \text{Cu}_{(\text{s})}; E^{\theta} = -0.34\text{V}$
- (a) Write the cell representation for the cell made up of two half cells. (1mk)
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- (b) Write the cell equation for the cell reaction. (1mk)
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(c) Calculate the E^θ value for the cell. (1mk)

29. State the function of each of the following in the solvay process of production of Sodium Carbonate.

(a) Coke (1mk)

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(b) Cold water on the carbonation. (1mk)

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(c) Ammonia a generator. (1mk)

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30. (a) A student in form three was given two gases C_2H_6 and C_2H_4 . He added acidified Potassium Manganate (VII) to each solution. State the observations the student made. (2mks)

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(b) State one use of C_2H_4 . (1mk)

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